Data Visualization III

Download the Iris flower dataset or any other dataset into a DataFrame. Scan the dataset andgive the inference as:

1.List down the features and their types(ex, numeric, nominal) available in the dataset.
2.Create a histogram for each feature in the dataset to illustrate the feature distributions.
3.Create a boxplot for each feature in the dataset.
4.Compare distribution and identity outliers

```
#Loading libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
Iris = pd.read_csv('4.Iris.csv')
Iris
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Sp
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.4	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	v
146	147	6.3	2.5	5.0	1.9	v

```
Iris.shape
```

(150, 6)

```
Iris.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWi
count	150.000000	150.000000	150.000000	150.000000	150.0
mean	75.500000	5.843333	3.054000	3.758667	1.1
std	43.445368	0.828066	0.433594	1.764420	0.7
min	1.000000	4.300000	2.000000	1.000000	0.1
25%	38.250000	5.100000	2.800000	1.600000	0.3
50%	75.500000	5.800000	3.000000	4.350000	1.3
75%	112.750000	6.400000	3.300000	5.100000	1.8
max	150.000000	7.900000	4.400000	6.900000	2.5

```
Iris.dtypes
```

	0
Id	int64
SepalLengthCm	float64
SepalWidthCm	float64
PetalLengthCm	float64
PetalWidthCm	float64
Species	object

dtype: object

```
Iris.isnull().sum()
```

	0
Id	0
SepalLengthCm	0
SepalWidthCm	0
PetalLengthCm	0
PetalWidthCm	0
Species	0

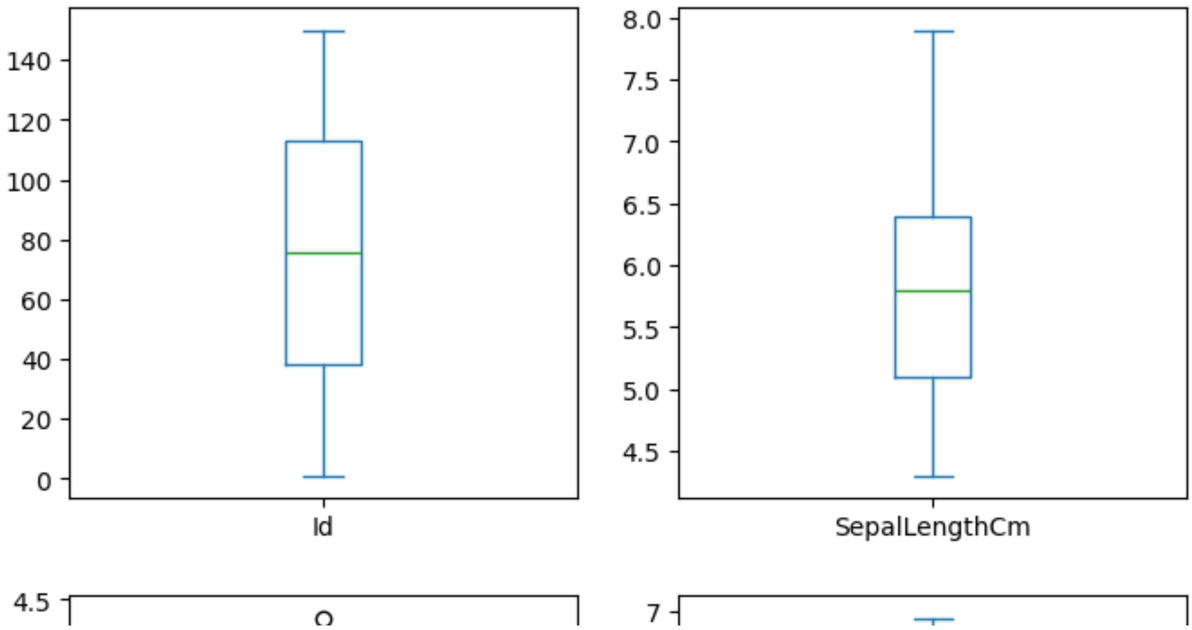
dtype: int64

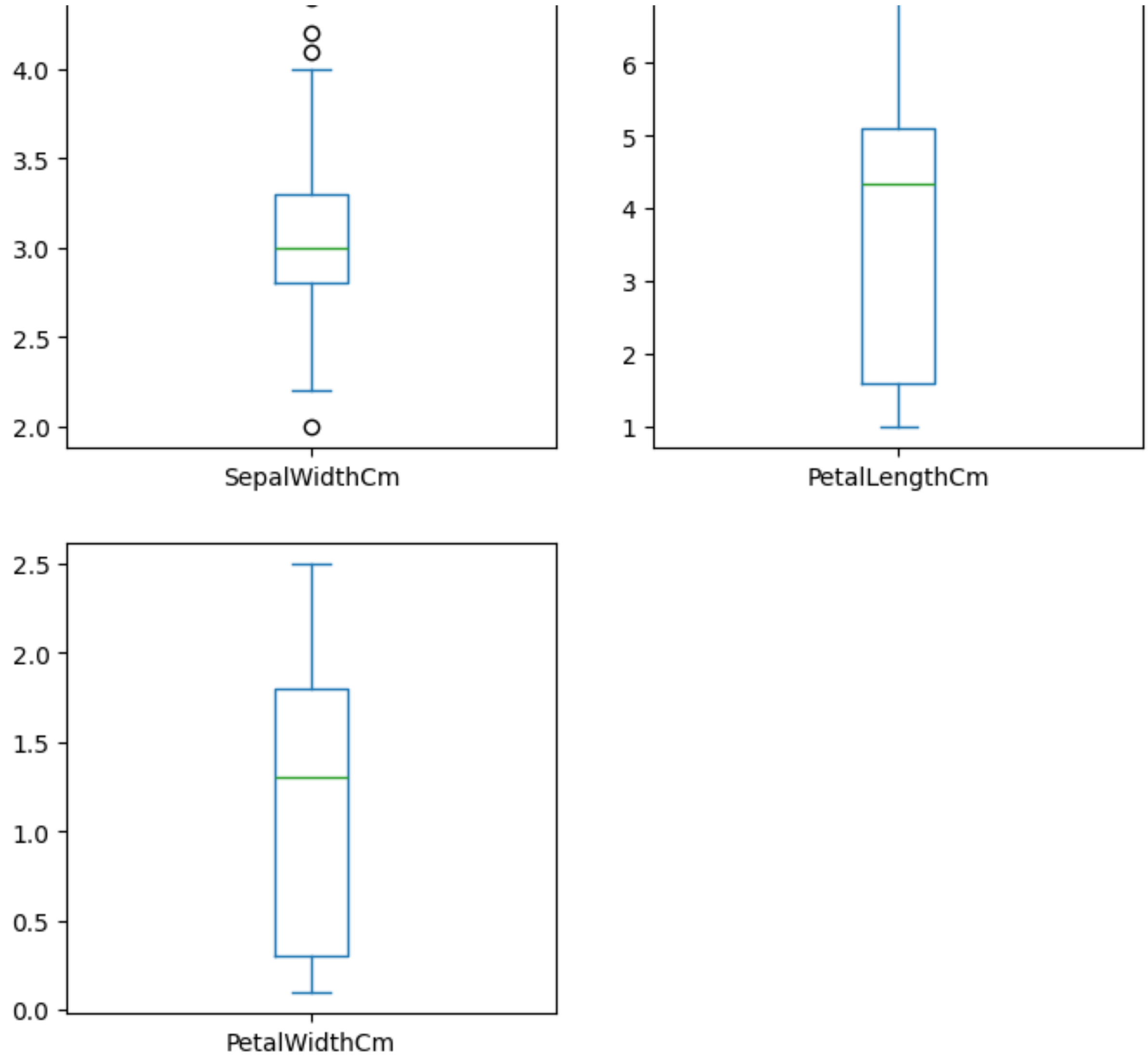
```
print(Iris.groupby('Species').size())
```

```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
dtype: int64
```

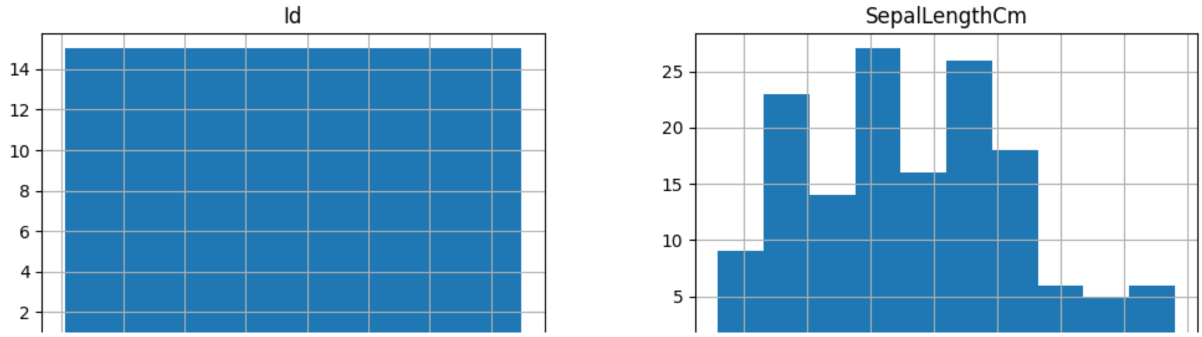
Data Visualization

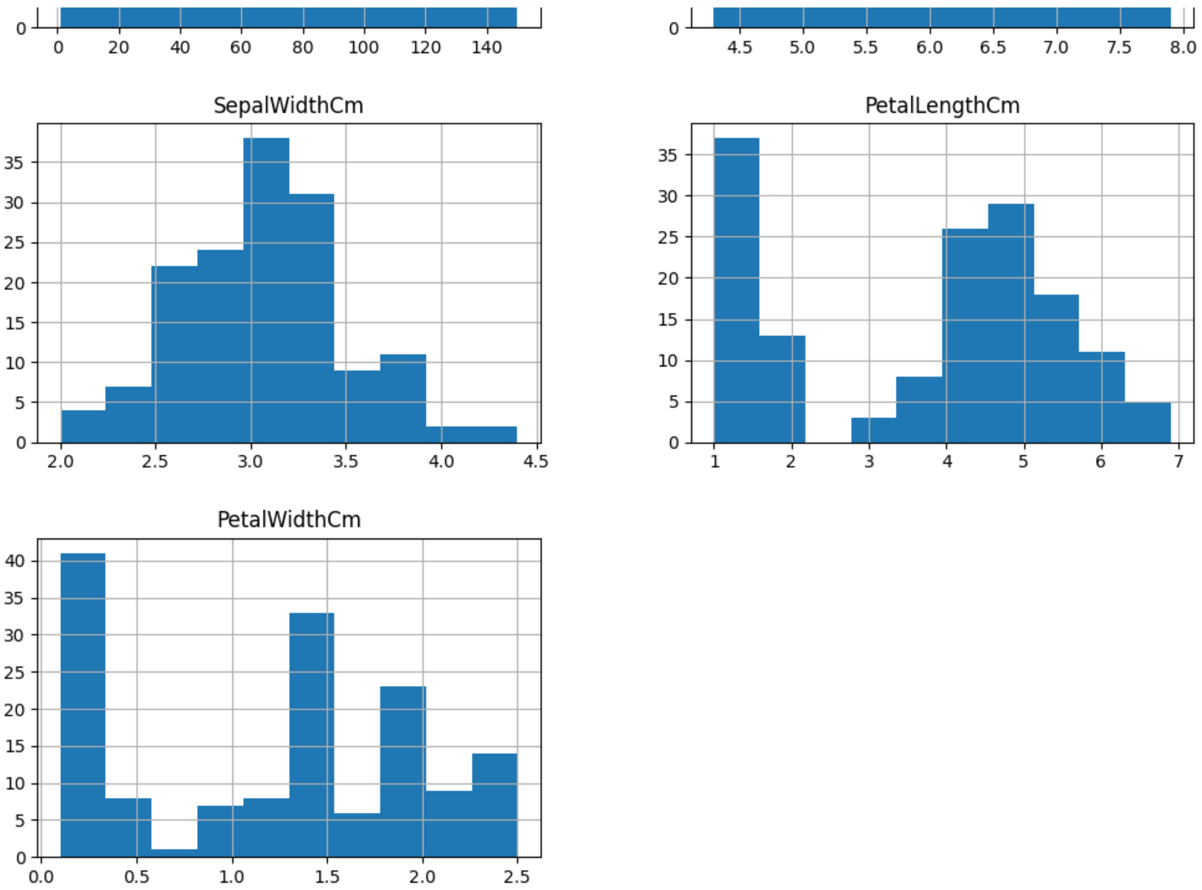
```
# Box and whisker plots
Iris.plot(kind='box', subplots=True, layout=(3,2), figsize=(8,12));
```






```
Iris.hist(figsize=(12,12))
plt.show()
```



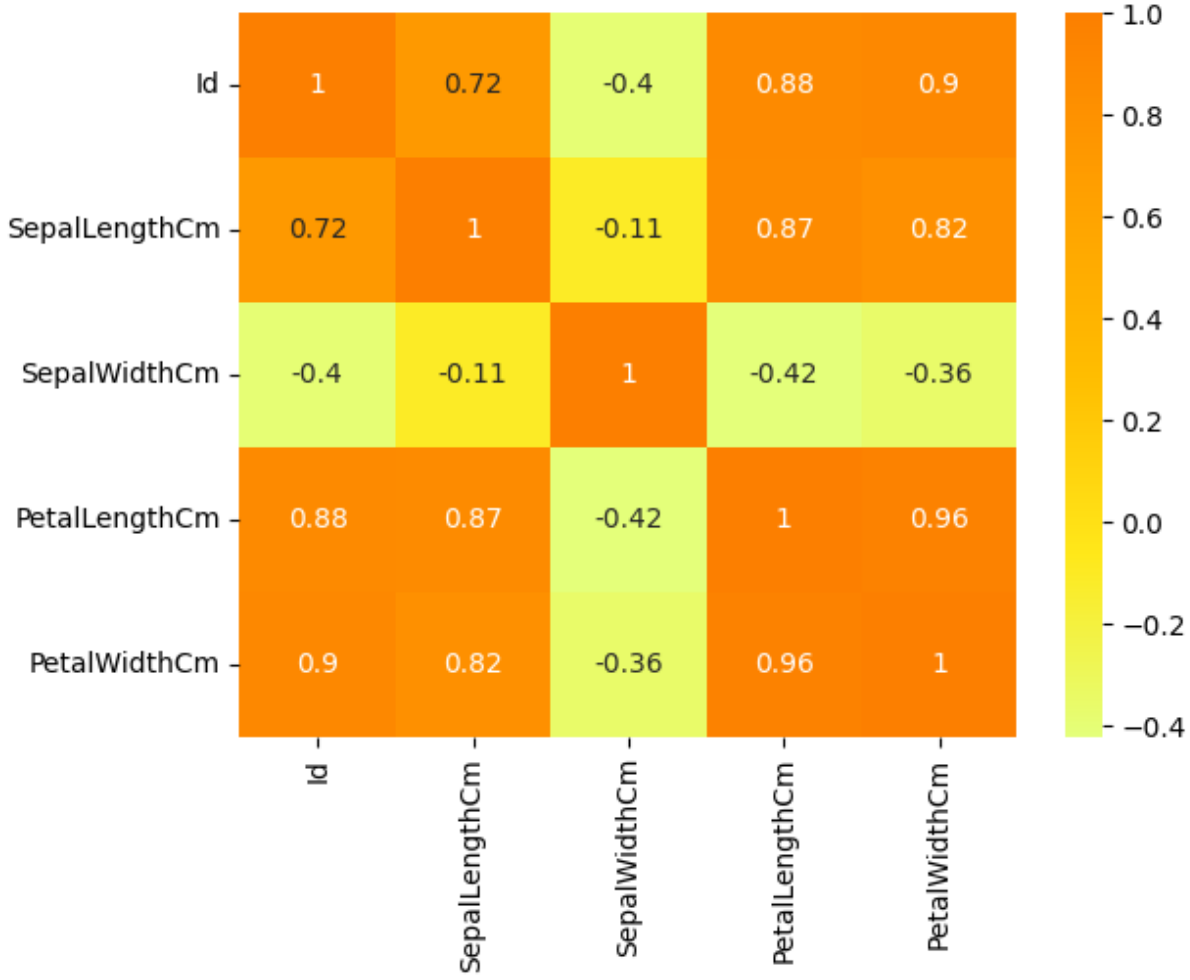

SepalWidthCm
PetalLengthCm
PetalWidthCm

```
Iris.corr(numeric_only=True)
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
Id	1.000000	0.716676	-0.397729	0.882747
SepalLengthCm	0.716676	1.000000	-0.109369	0.871754
SepalWidthCm	-0.397729	-0.109369	1.000000	-0.420516
PetalLengthCm	0.882747	0.871754	-0.420516	1.000000
PetalWidthCm	0.899759	0.817954	-0.356544	0.962757

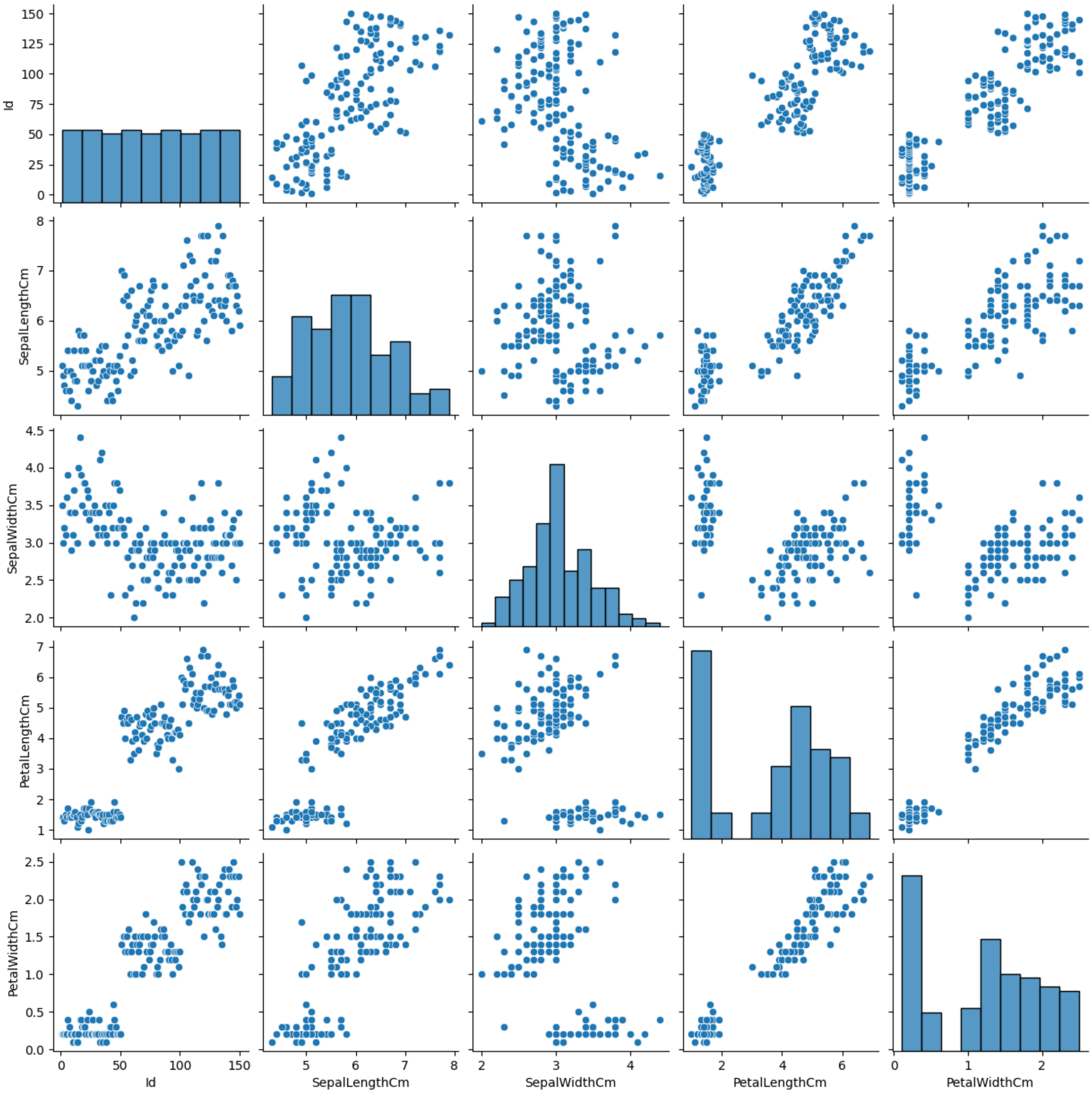
```
sns.heatmap(Iris.corr(numeric_only=True), annot=True, cmap='Wistia'
```

<Axes: >




```
sns.pairplot(Iris)
```

<seaborn.axisgrid.PairGrid at 0x7ce6174a8f80>


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
0 25 50 75 100 125 150
4 5 6 7 8
2.0 2.5 3.0 3.5 4.0 4.5
1 2 3 4 5 6 7
0.0 0.5 1.0 1.5 2.0 2.5
0 50 100 150
5 6 7 8
2 3 4
2 4 6
0 1 2

Step 1: Load the dataset into a DataFrame

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
iris_df = pd.DataFrame(iris.data, columns=iris.feature_names)
iris_df['target'] = iris.target
```

Step 2: List down the features and their types

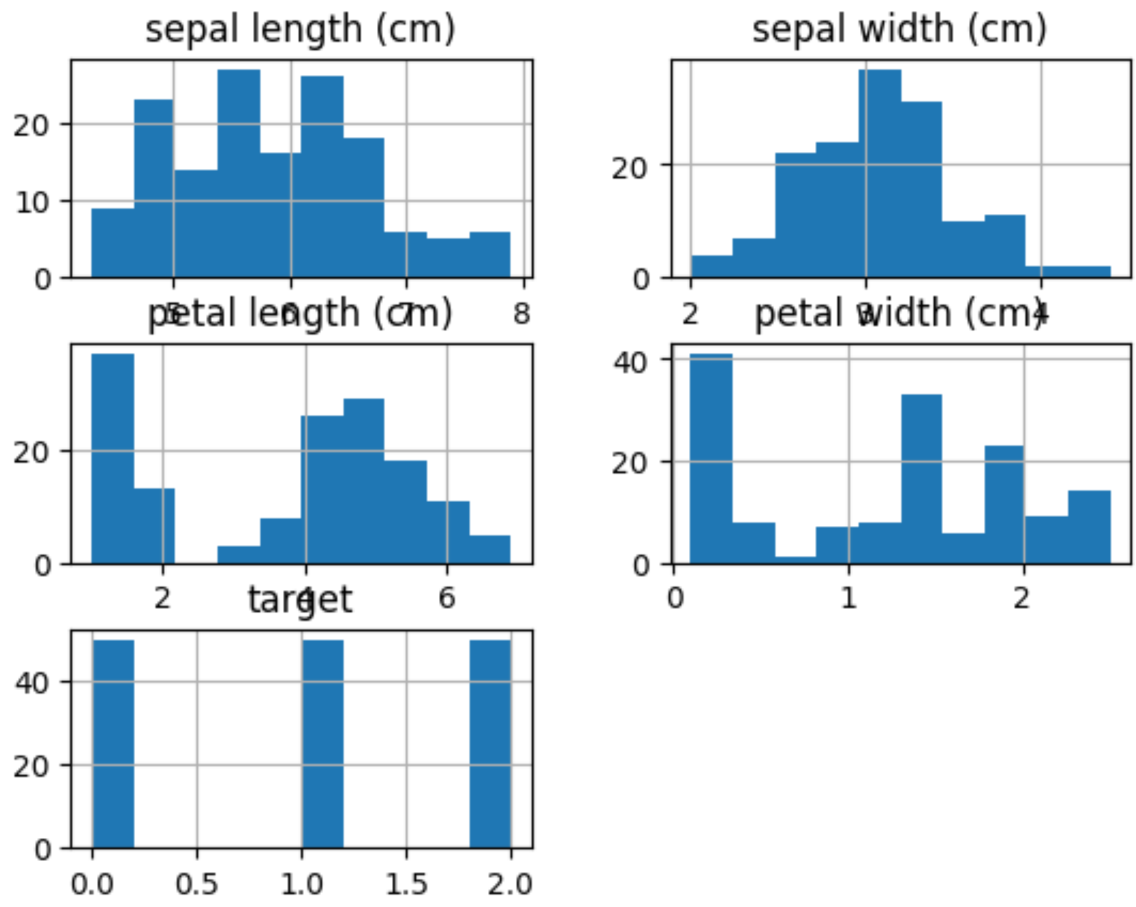
The Iris flower dataset contains the following features:

- sepal length(numeric)
- sepal width (numeric)
- petal length(numeric)
- petal width (numeric)
- target (nominal)

Step 3: Create a histogram for each feature

To create a histogram for each feature, we can use the hist method of the DataFrame as follows:

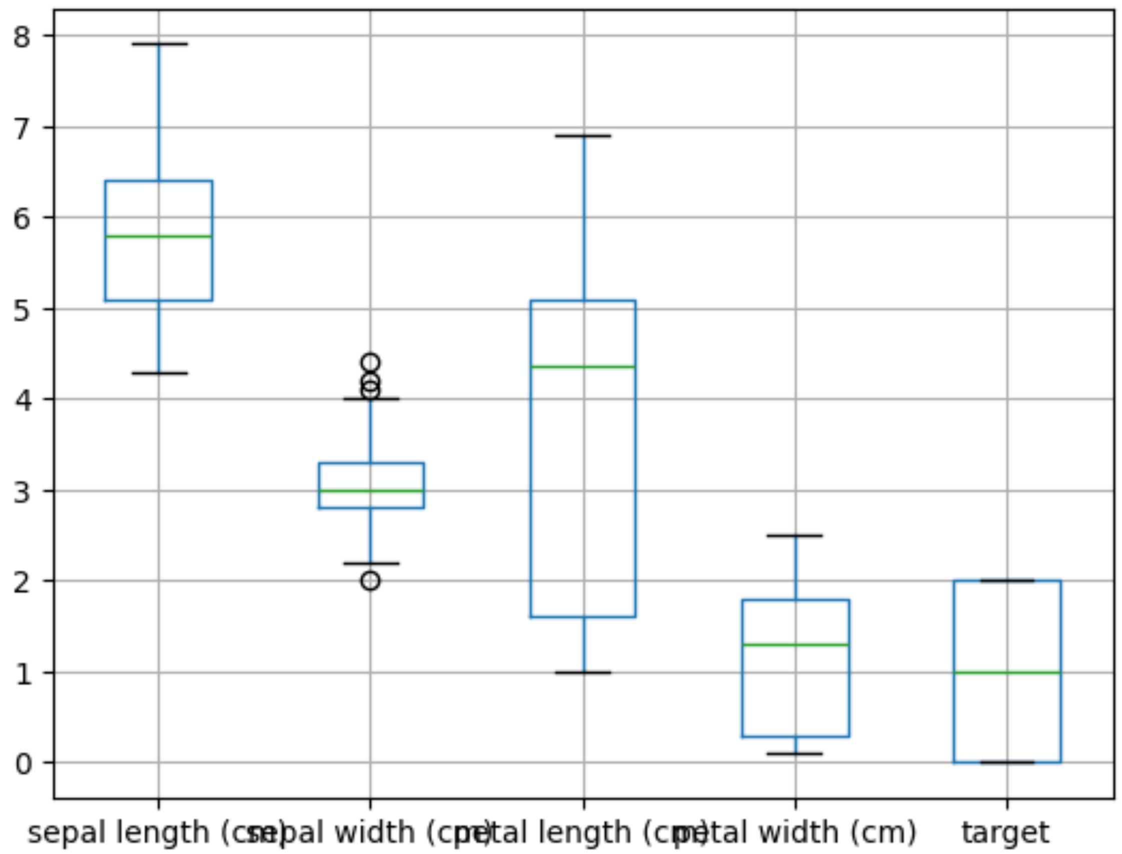
```
import matplotlib.pyplot as plt

iris_df.hist()
plt.show()
```



Step 4: Create a boxplot for each feature

To create a boxplot for each feature, we can use the boxplot method of the DataFrame as follows:

```
iris_df.boxplot()
plt.show()
```



Step 5: Compare distribution and identify outliers By looking at the histograms and boxplots, we can see the distribution of each feature and identify any outliers.

For example, we can see that the petal length and petal width have a bimodal distribution. The sepal length and sepal width have a more normal distribution. We can also see that there are some outliers in the sepal width and petal length features.

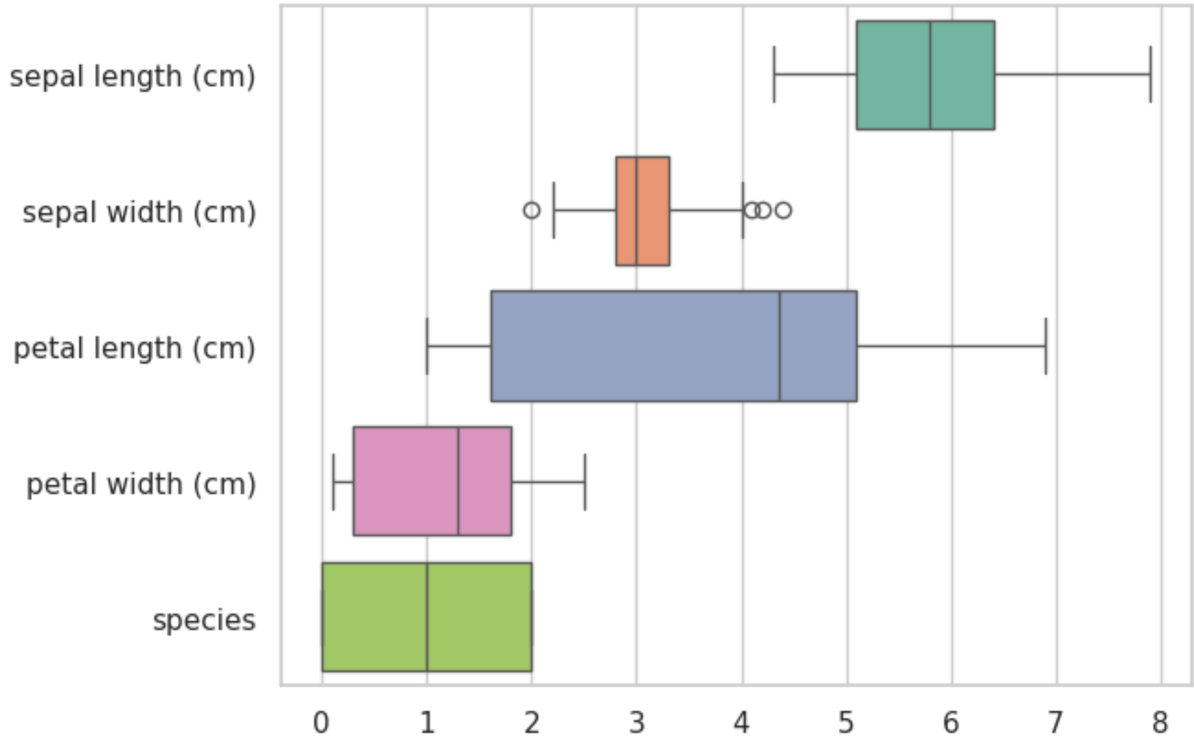
We can further investigate the outliers using the describe method of the DataFrame as follows:

```
iris_df.describe()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

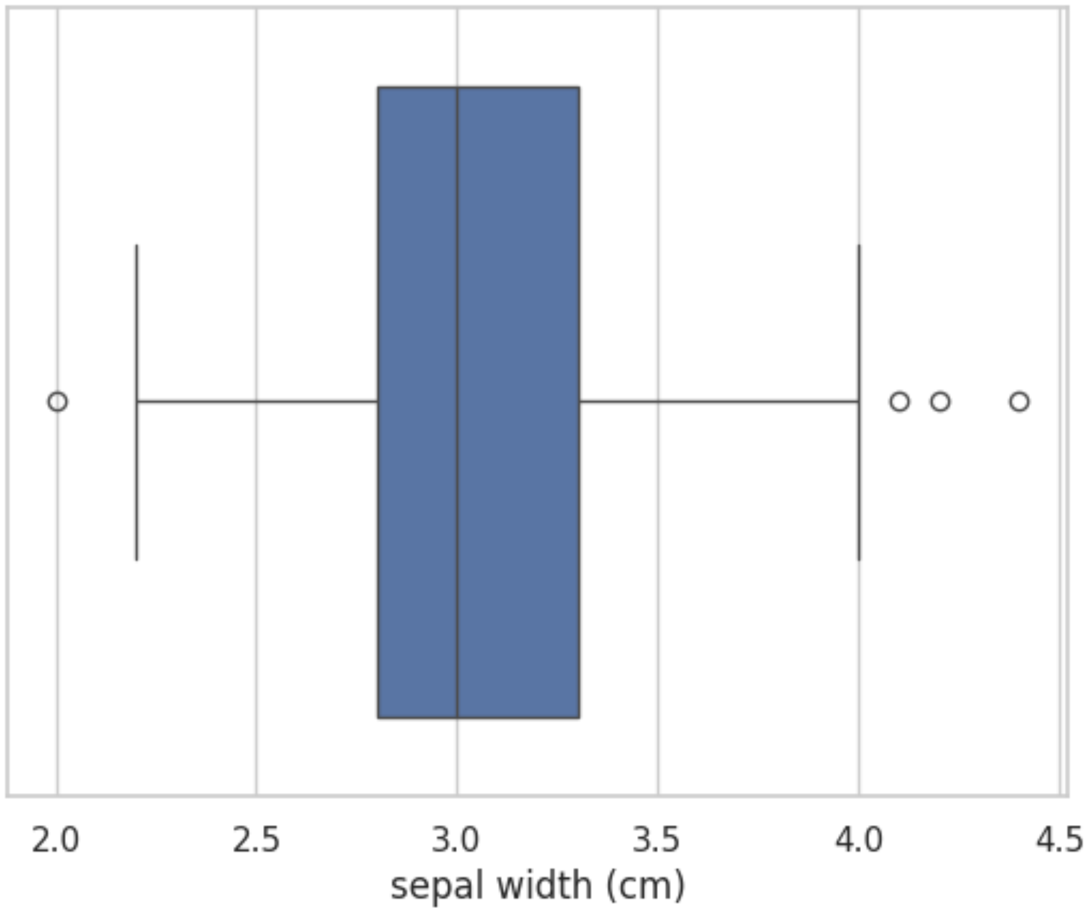
```
import pandas as pd
from sklearn.datasets import load_iris

iris = load_iris()

# Convert to DataFrame
df = pd.DataFrame(iris.data, columns=[
    'sepal_length', 'sepal_width', 'petal_length', 'petal_width'
])
```

```
sns.set(style="whitegrid")
sns.boxplot(data=df, orient="h", palette="Set2")
plt.show()
```



```
sns.boxplot(x = 'sepal width (cm)', data = iris_df)
```

```
<Axes: xlabel='sepal width (cm)'>
```




```
Q1 = Iris.SepalWidthCm.quantile(0.25)
Q3 = Iris.SepalWidthCm.quantile(0.75)
IQR = Q3-Q1
print(IQR)
```

```
0.5
```

```
data = Iris[Iris.SepalWidthCm < (Q1 - 1.5 * IQR) / (Iris.SepalWidtl
```

```
data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Sp
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.4	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	v
146	147	6.3	2.5	5.0	1.9	v